TACS 25/26 ASSIGNMENT

- Alexandre Cunha - up202400086
- Artur Oliveira - up202108663
- Eduardo Areias - up202410456
- Ricardo Peralta - up202206392
- Rodrigo Póvoa - up202108890
- Rodrigo Moucho - up202108855

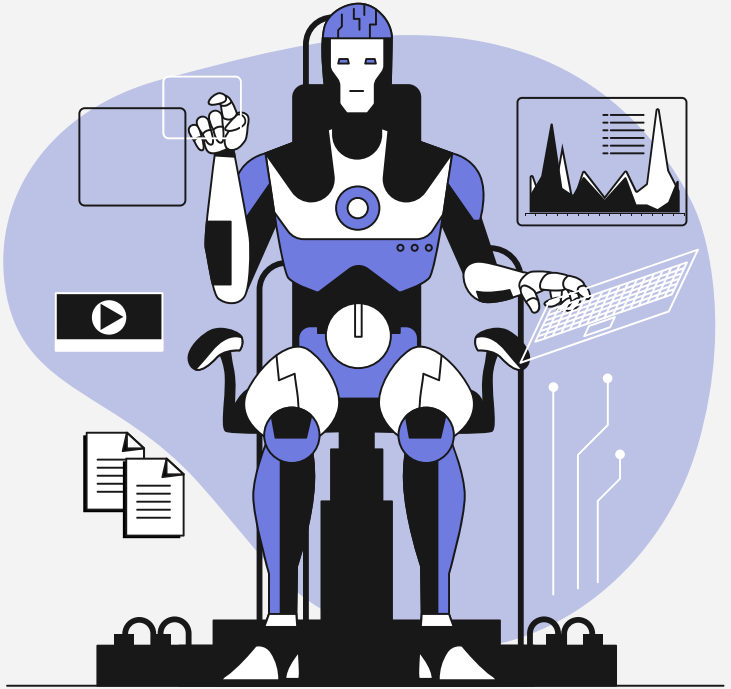


Table of contents

01 Structural Metamodel

02 DSML Metamodel

03 GUI Metamodel

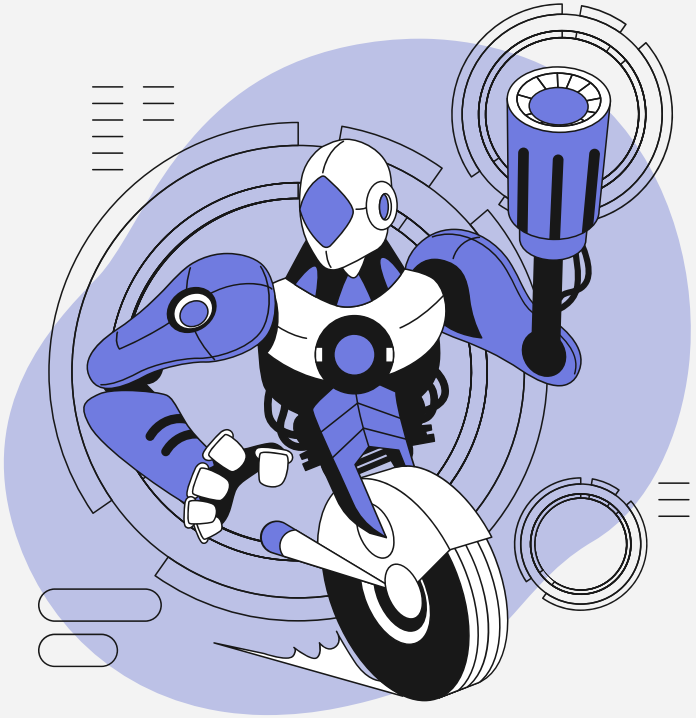
04 Model to Model

05 Model to Text

06 Final Result

App to Exchange/Sell/Auction Items

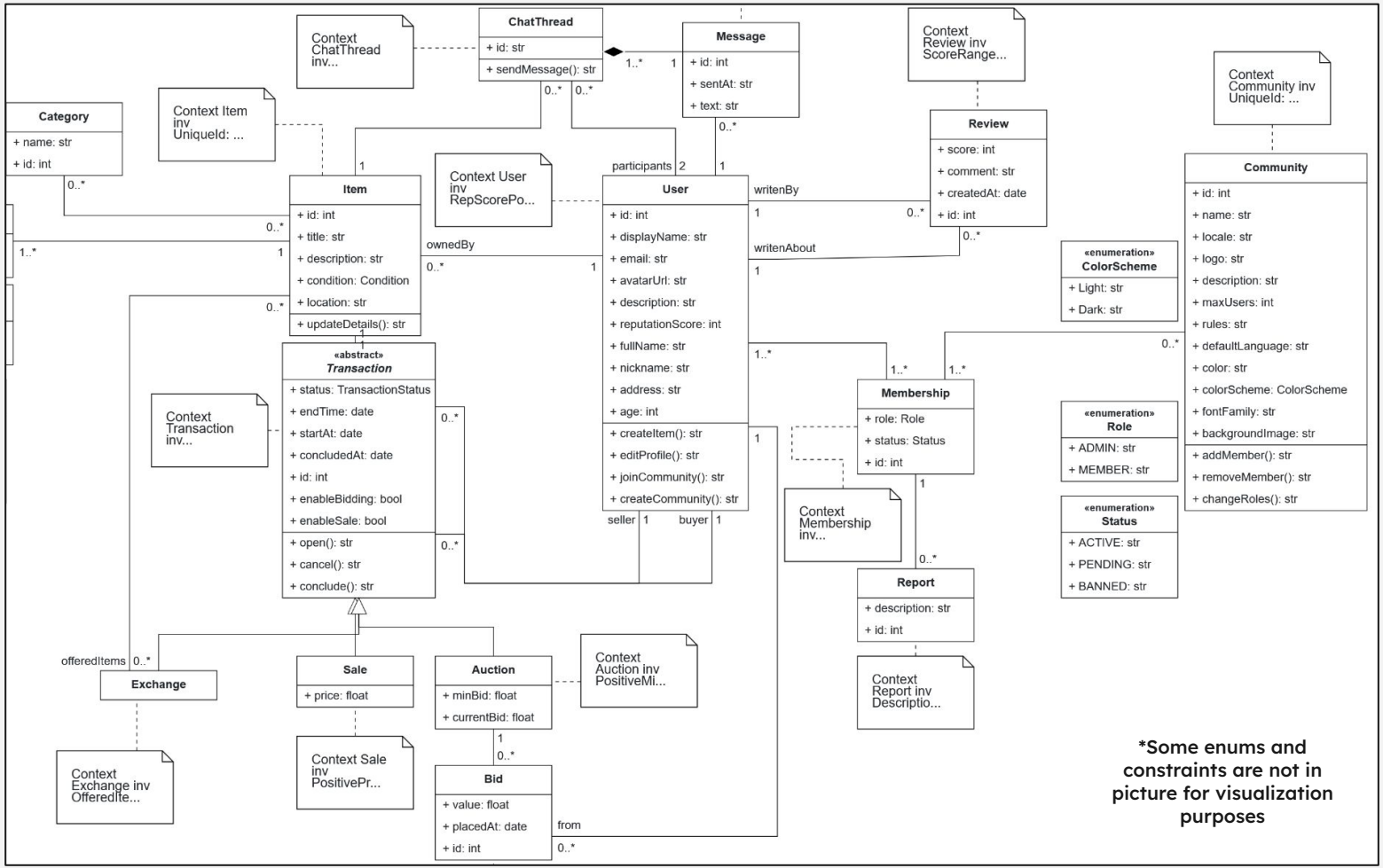
- **BESSER** framework to create structural, DSML and GUI metamodels
- Model2Model, using the **B-UML** and **Blocky**
  - Structural and GUI models
- Model to Text, using **Jinja**
  - ChatScreen, ListingDetailsScreen, CreateListingScreen



01

Structural Metamodel

Structural Metamodel

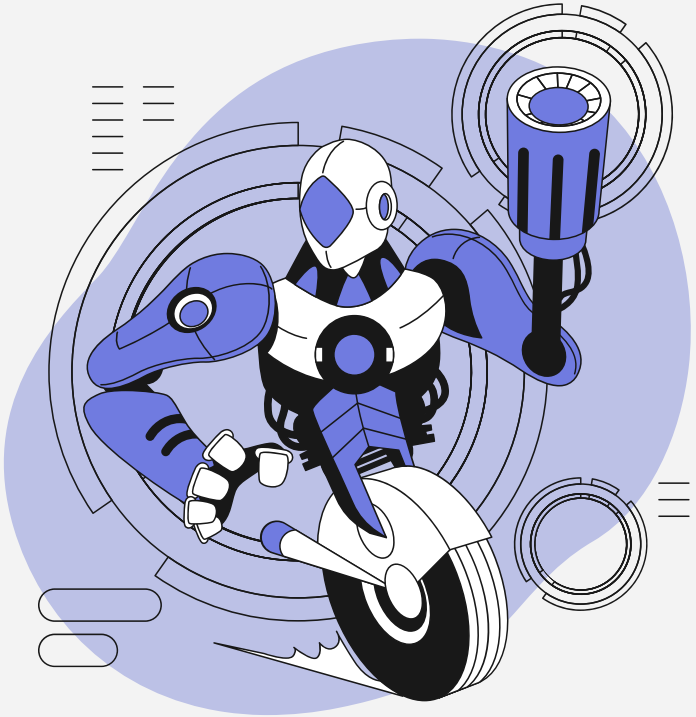


*Some enums and constraints are not in picture for visualization purposes

Structural Metamodel

Constraints include:

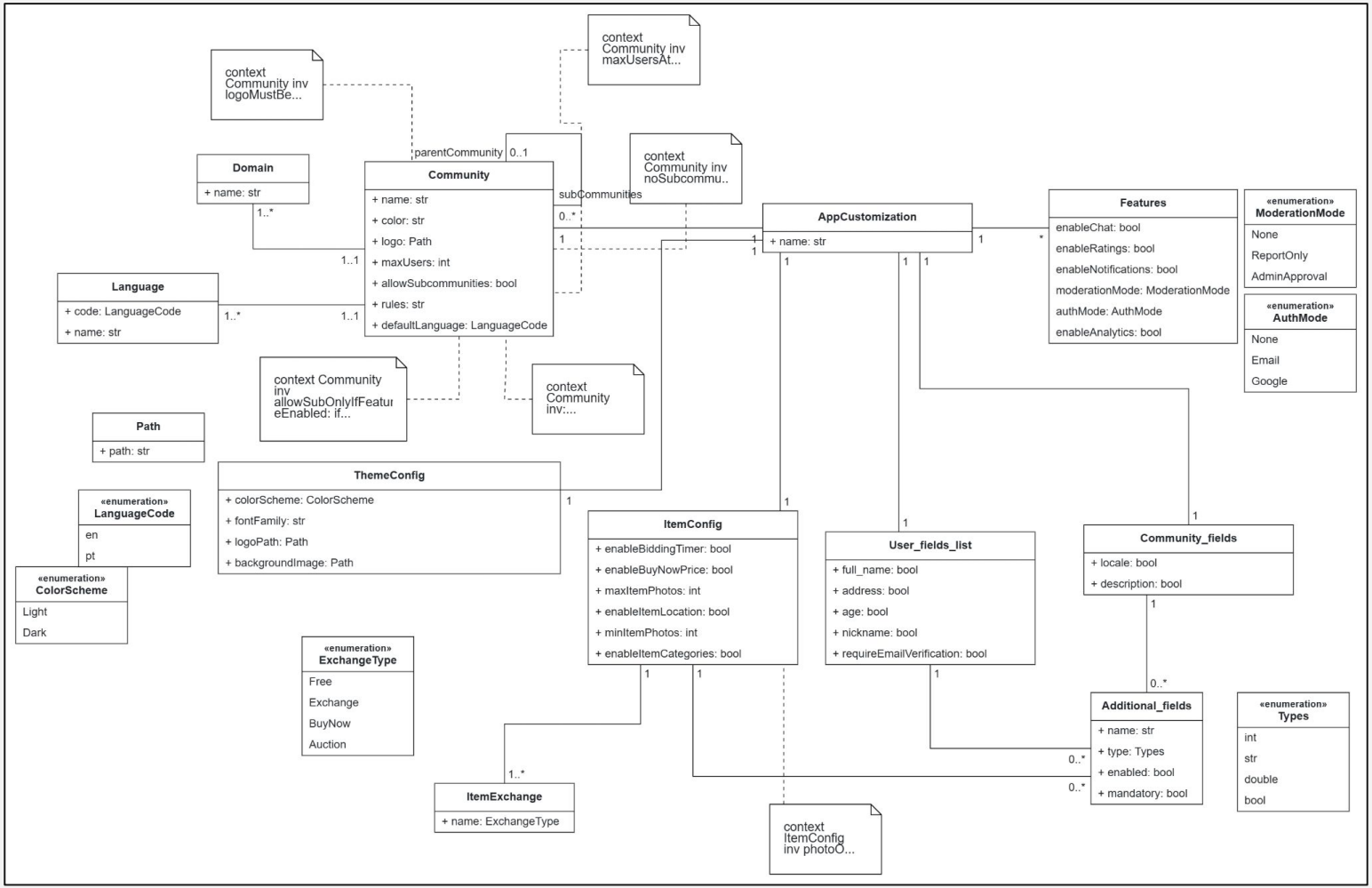
- Item
  - Must have a title, description, category and at least one photo
- User
  - Must have a unique id and email address and a reputation score between 0.0 and 5.0
- Community
  - Must have an unique name and description under 2000 characters



02

DSML Metamodel

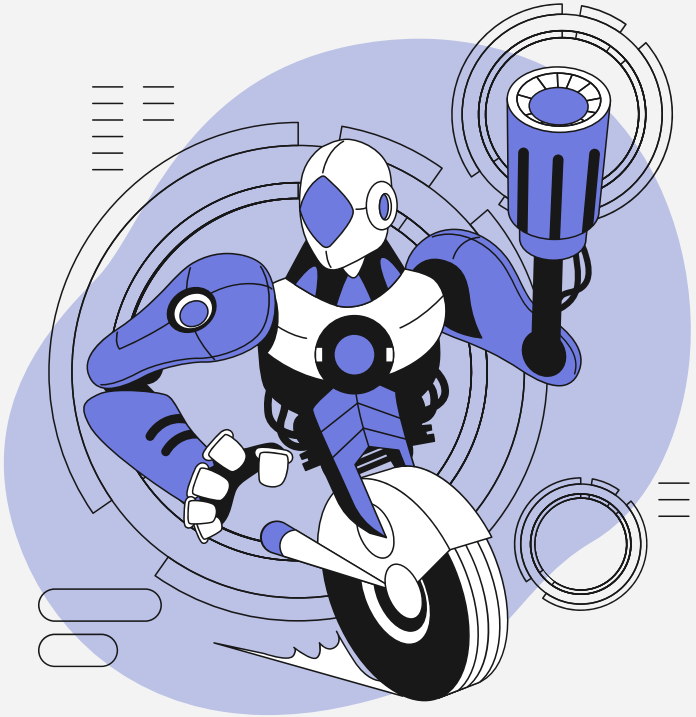
DSML Metamodel



DSML Metamodel

Constraints include:

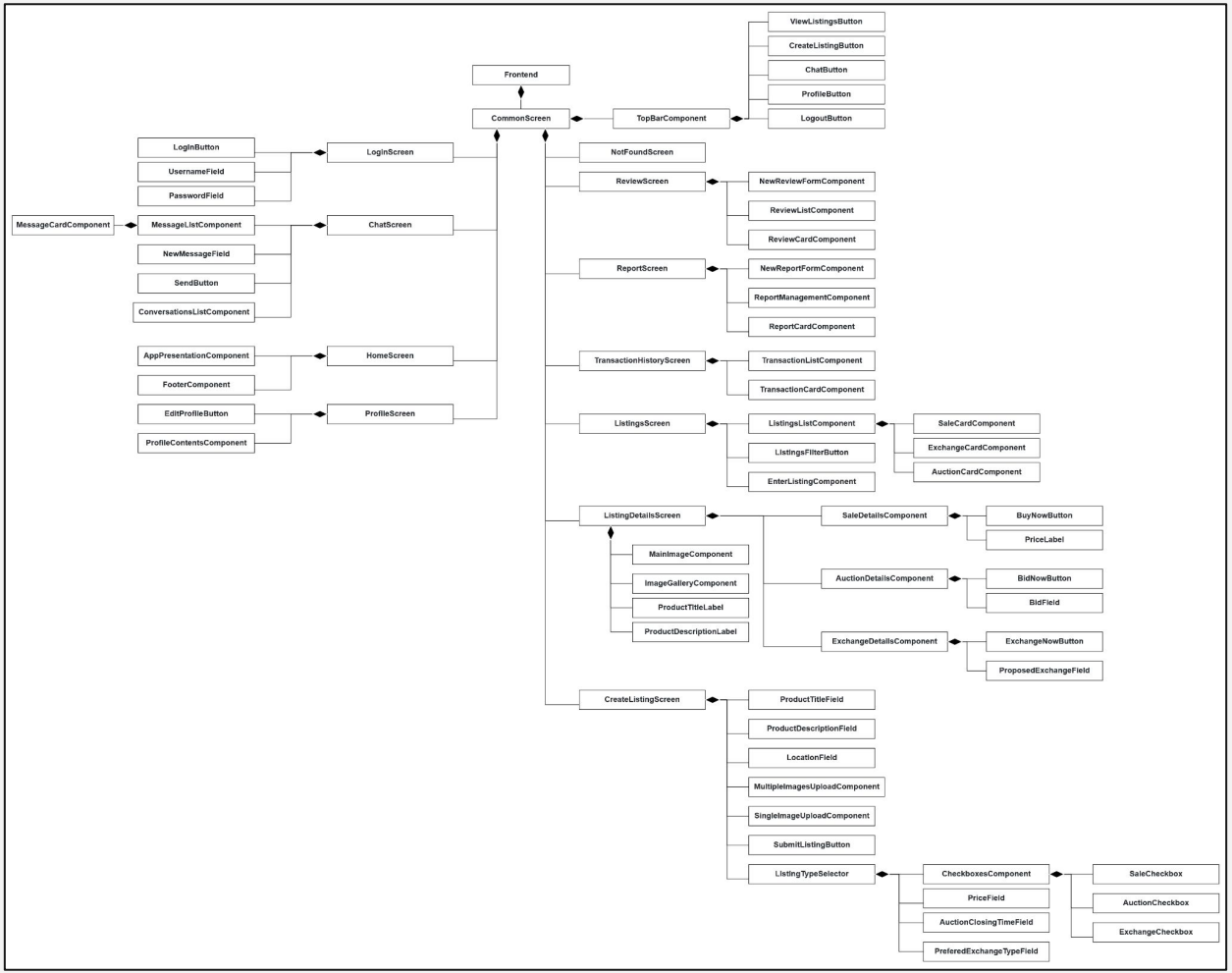
- A community that does not allow subcommunities must not contain any subcommunities
- A community cannot reference itself as its own parent community
- A community’s logo file must be an image, with a file extension of png, jpg, jpeg, gif, or svg



03

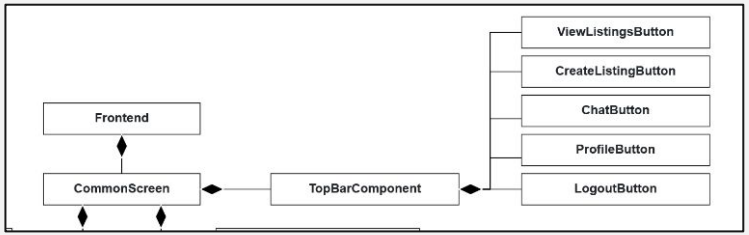
GUI Metamodel

GUI Metamodel



GUI Metamodel

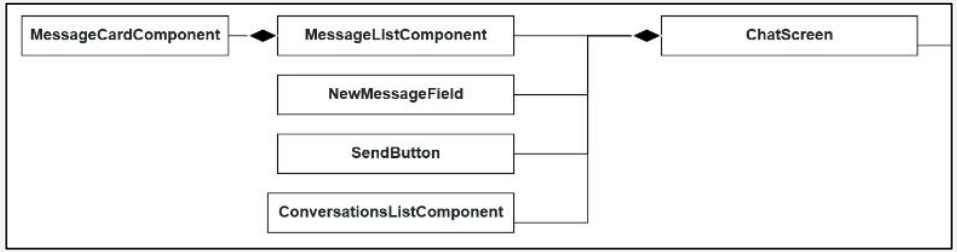
Common to all pages



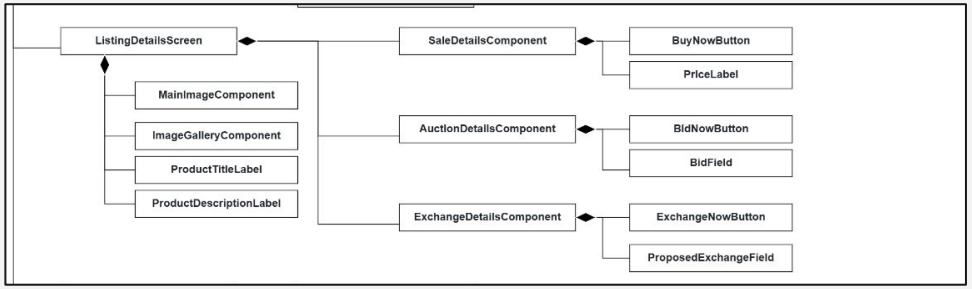
- LoginScreen
- ListingsScreen
- HomeScreen
- ProfileScreen
- ReviewScreen
- ReportScreen
- TransactionHistoryScreen
- **ChatScreen**
- **ListingDetailsScreen**
- **CreateListingScreen**

GUI Metamodel Pages to be Implemented

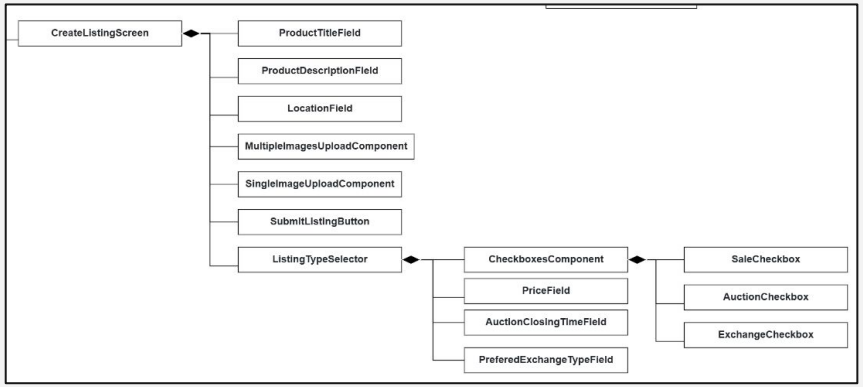
ChatScreen

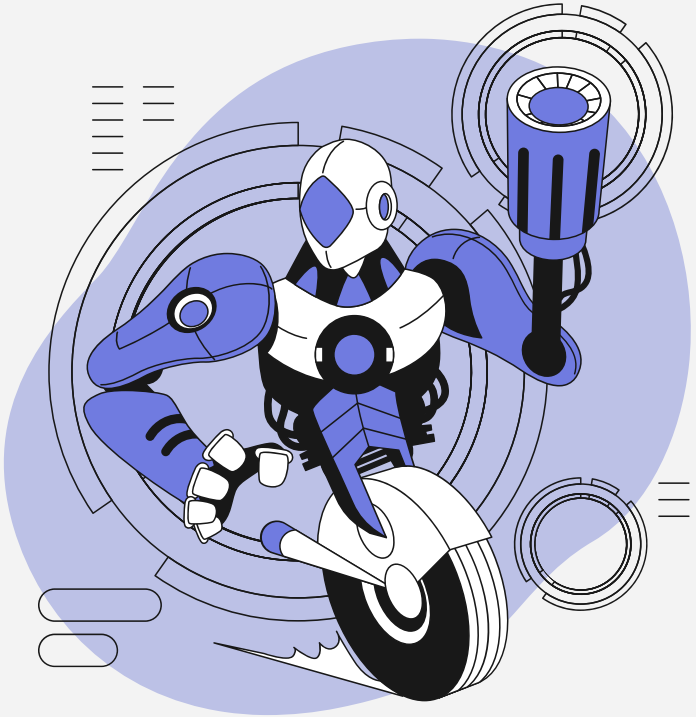


ListingDetailsScreen



CreateListingScreen





04

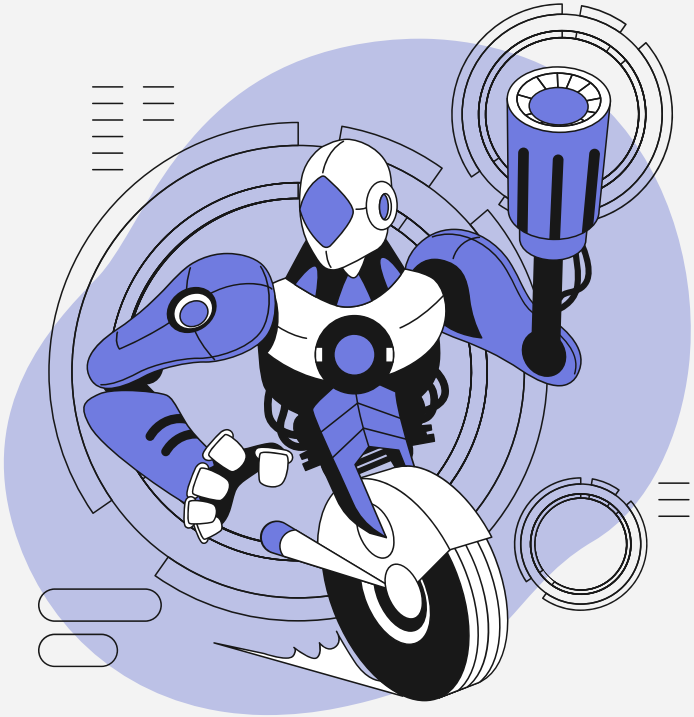
Model to Model

Model to Model

- Export the **BESSER** structural and GUI metamodel to B-UML
- Export the DSML metamodel to json and use **Blocky** to create the model and export to json
- Create a python script that consumes the B-UML and json and generates a **domain model** for both

Blocky


AppCustomization name Teste123 itemconfig
ItemConfig enableBiddingTimer True enableBuyNowPrice False maxItemPhotos 5 enableItemLocation True
minItemPhotos enableItemCategories appcustomization
additional_fields
itemexchange
ItemExchange name Auction itemconfig
ItemExchange name Free itemconfig
features
Features enableChat True enableRatings False enableNotifications True moderationMode None
authMode None enableAnalytics True appcustomization
community
Community name Example community color #0000ff logo https://picsum.photos/200/300 maxUsers 10
allowSubcommunities False rules No disrespect allowed. defaultLanguage en appcustomization
subCommunities
parentCommunity
language
domain
Domain name Feup community
themeconfig
ThemeConfig colorScheme Light fontFamily Arial logoPath https://picsum.photos/200/300 backgroundImage https://picsum.photos/200/300
appcustomization
user_fields_list
User_fields_list full_name True address True age True nickname True
requireEmailVerification appcustomization
additional_fields
community_fields
Community_fields locale True description True appcustomization
additional_fields

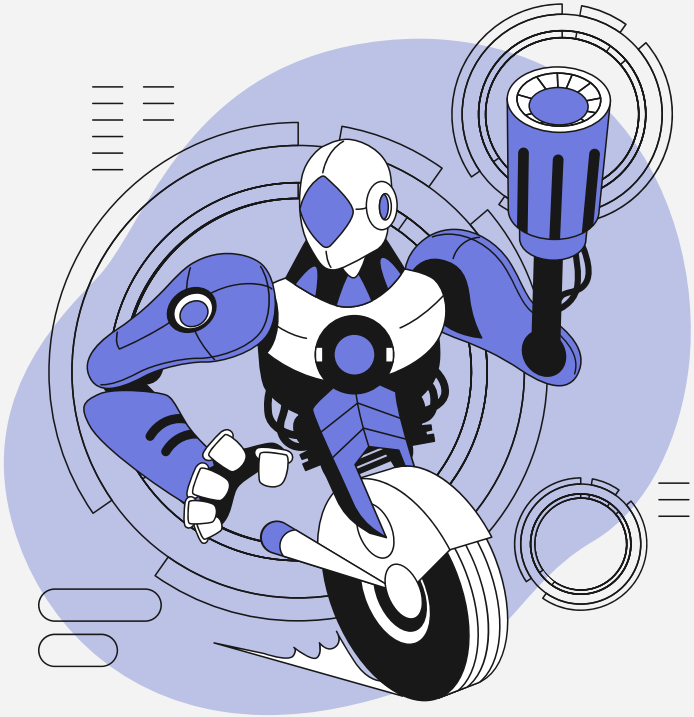


05

Model to Text

Model to Text

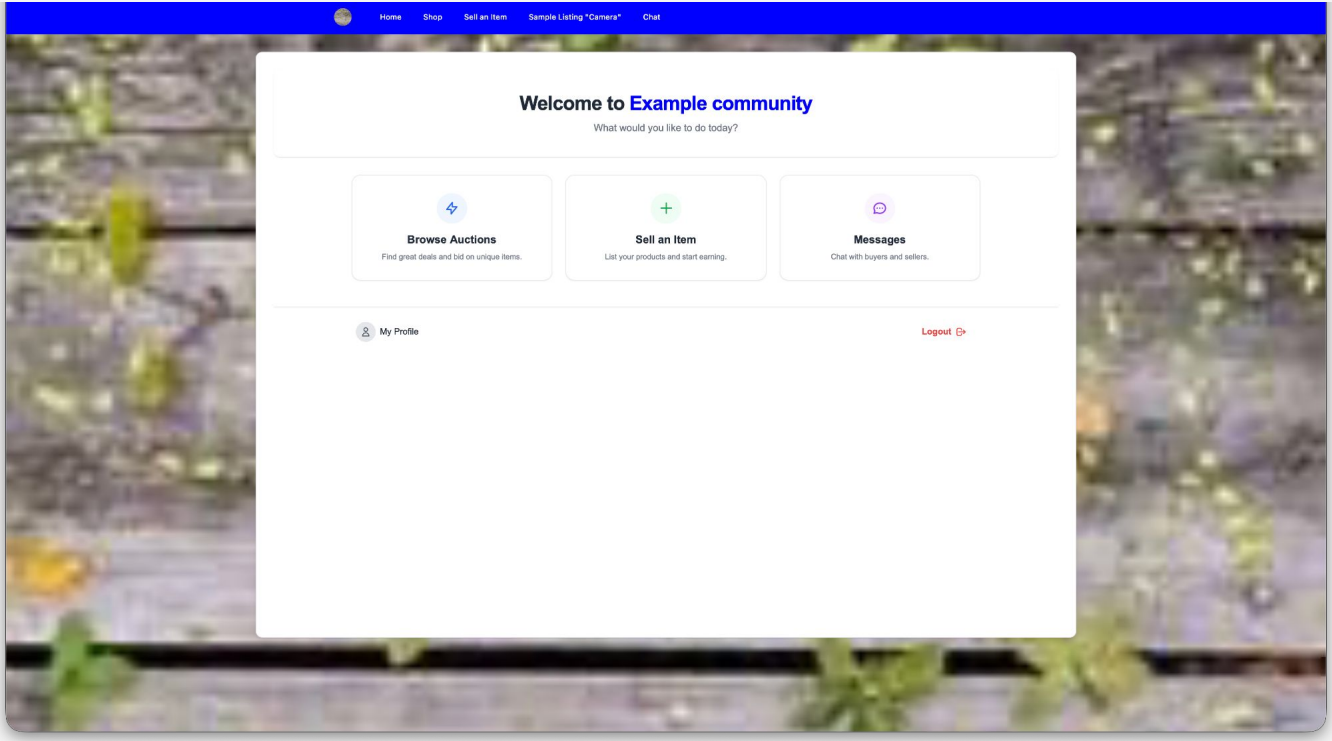
- **DSML config** not existent in GUI model to json
- Feed to generator that implements **GeneratorInterface**
- **GeneratorInterface** receives **GUI model**, **DSML config** and **Jinja Templates**
- Create a **JinjaContext** to render the **selected pages**



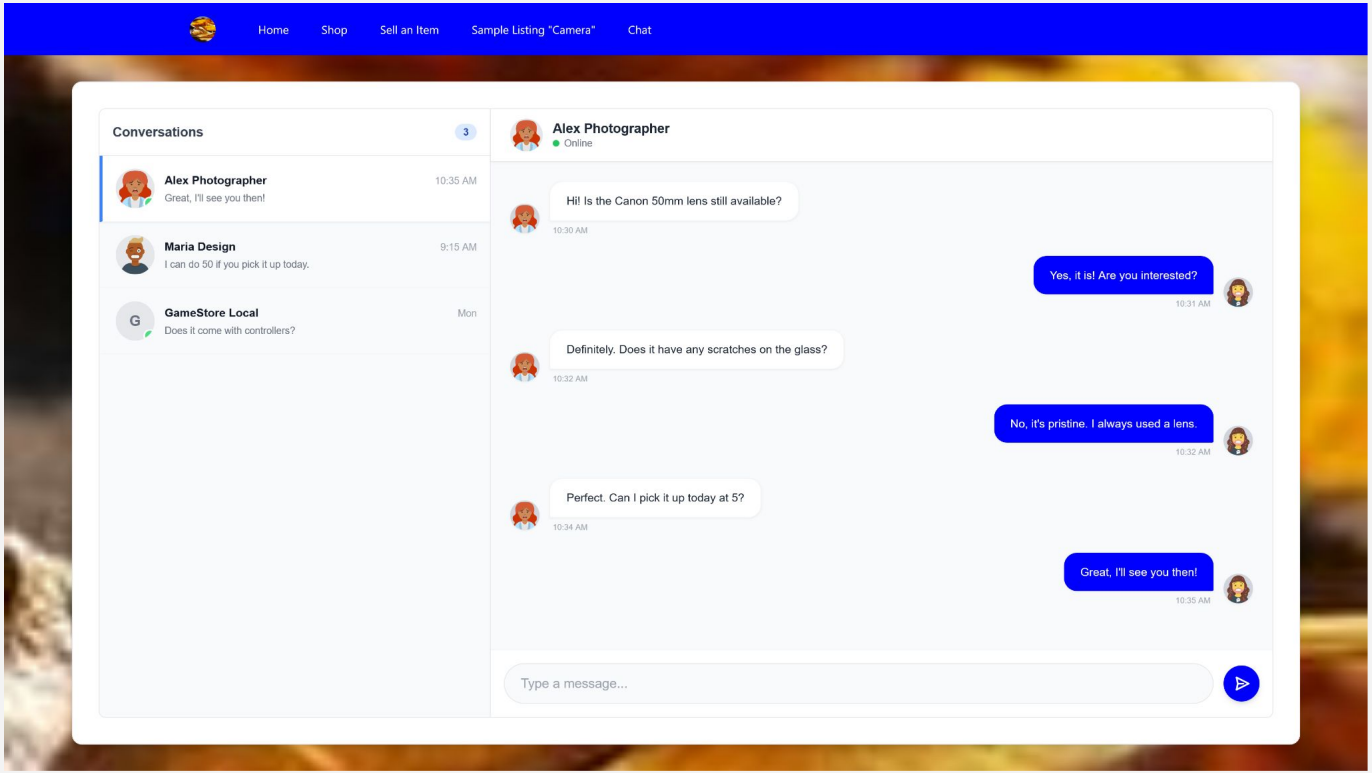
06

Final Result

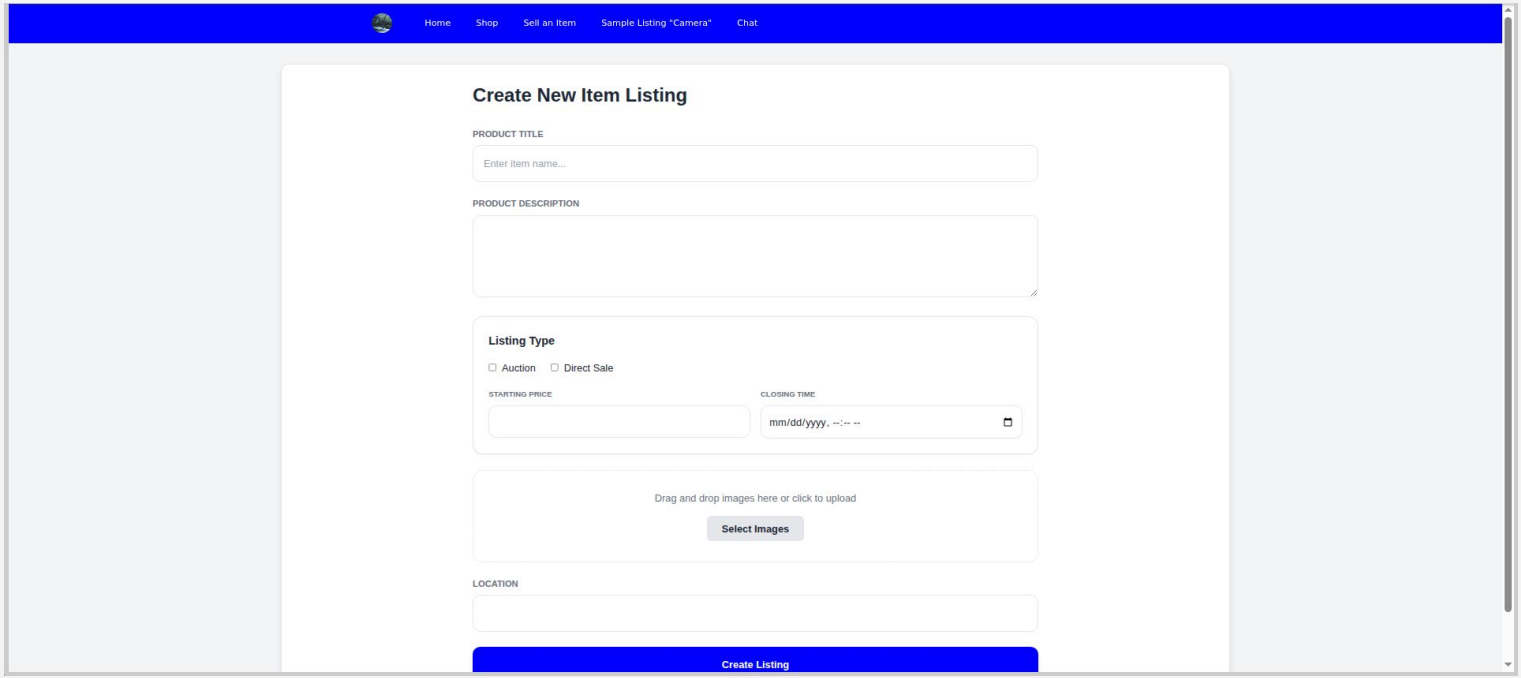
HomeScreen


Home
Shop
Sell an Item
Sample Listing "Camera"
Chat
Welcome to Example community
What would you like to do today?
Browse Auctions
Find great deals and bid on unique items.
Sell an Item
List your products and start earning.
Messages
Chat with buyers and sellers.
My Profile
Logout

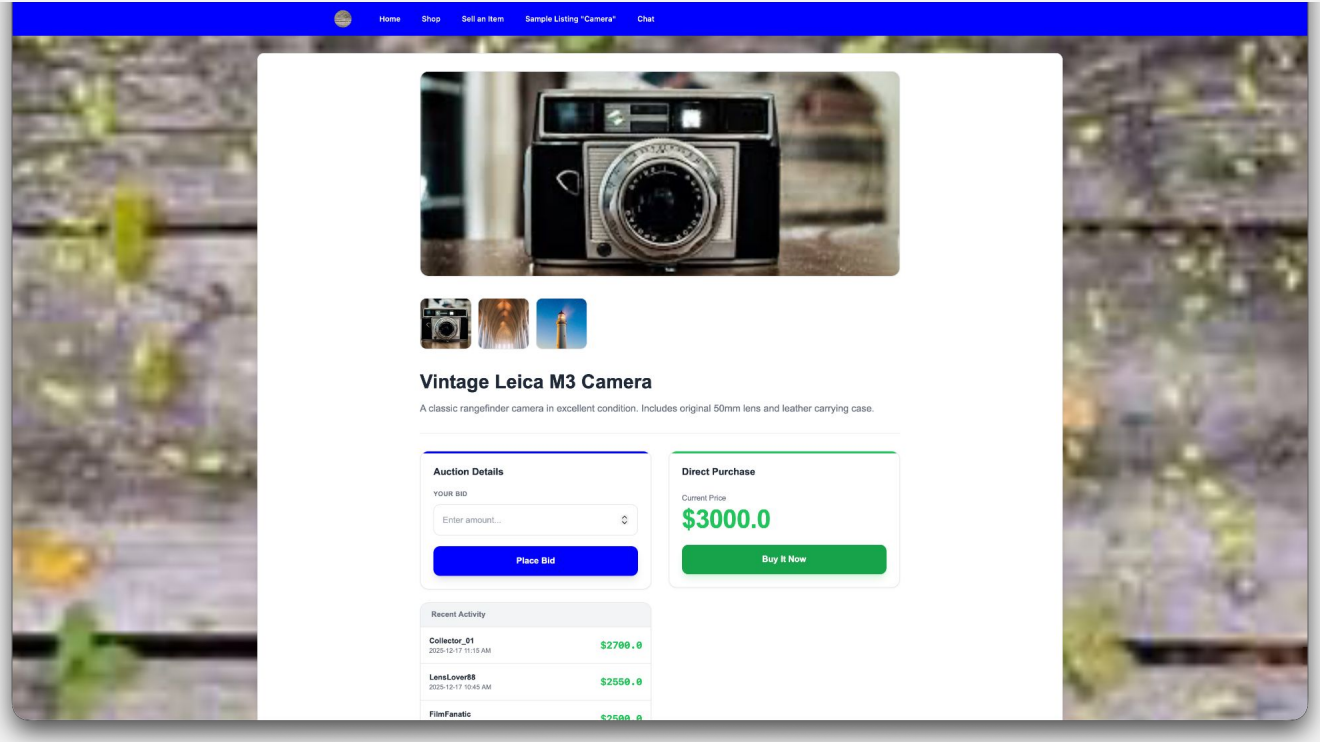
ChatScreen

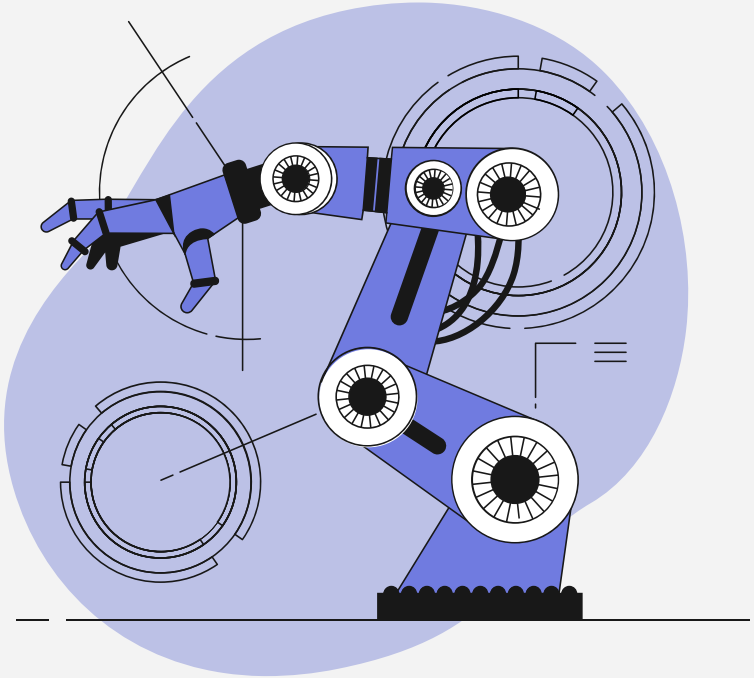

Home
Shop
Sell an Item
Sample Listing "Camera"
Chat
Conversations
3
Alex Photographer
Great, I'll see you then!
10:35 AM
Maria Design
I can do 50 if you pick it up today.
9:15 AM
G
GameStore Local
Does it come with controllers?
Mon
Alex Photographer
Online
Hi! Is the Canon 50mm lens still available?
10:30 AM
Yes, it is! Are you interested?
10:31 AM
Definitely. Does it have any scratches on the glass?
10:32 AM
No, it's pristine. I always used a lens.
10:32 AM
Perfect. Can I pick it up today at 5?
10:34 AM
Great, I'll see you then!
10:35 AM
Type a message...

CreateListingScreen



ListingDetailsScreen





Questions?